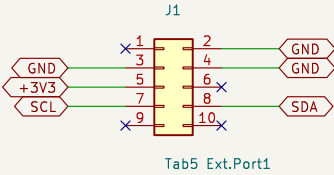
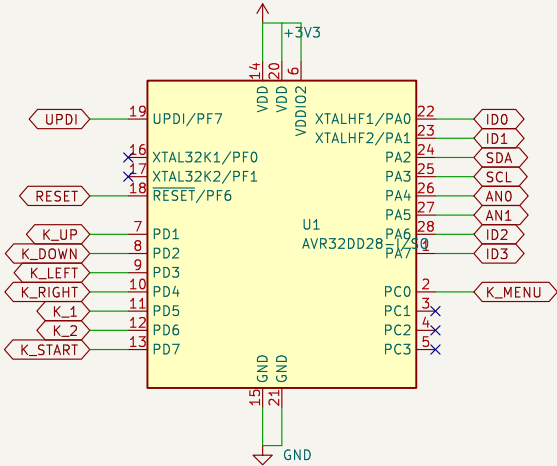


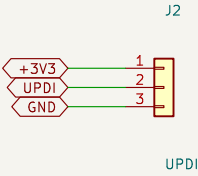
MCU – AVR32DD28 (SOIC–28) running firmware_avrdd (I2C client 0x6D, RetroPad Ext. Port 1) – verify pin 1 orientation before fab

Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware_avrdd/pinmap.h

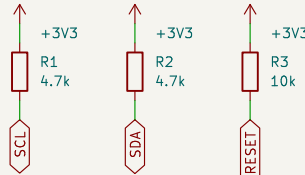
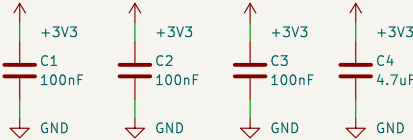
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



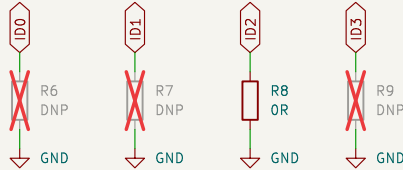
UPDI programming (SerialUPDI / MPLAB Snap)



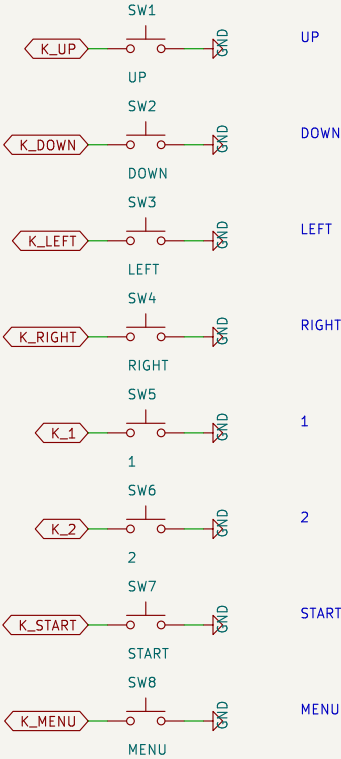
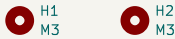
Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up



Console ID straps: OR to GND = bit set. Master System / Game Gear = ID 4 (fit R8; R6, R7, R9 DNP)



Mounting holes (M3, 96 mm apart)



Tab5 keyboard–port board: AVR32DD28 + firmware_avrdd, I2C 0x6D
Generated by kicad_gen/gen_sch.py from gen_pcb.py and layout.py
RetroESP32–P4
Sheet: /
File: retropad_sms.kicad_sch

Title: RetroPad Master System / Game Gear controller (console ID 4)

Size: A3	Date: 2026–10–05	Rev: 0.1
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4	Id: 1/1	